

Family Cimicidae

Bed bugs

Occurrence 90 spp. worldwide; on bird and mammal hosts, in nests, caves, crevices in buildings



These flat, oval, reddish brown bugs are $\frac{5}{32}$ – $\frac{3}{8}$ in (4–12 mm) long, and have remnant front wings and no hindwings. Their compound eyes have a small number of facets. All bed bugs are blood sucking, temporary ectoparasites that live on birds, bats, and even humans. They live for 2–10 months depending on feeding and temperature. Males practice traumatic insemination, injecting sperm by penetrating the female's body with a sharp sexual appendage. Females may lay more than 150 eggs in their adult life, with 2 or 3 eggs laid per day.



CIMEX LECTULARIUS
The common bed bug feeds at night and senses the presence of a host using temperature, odor, and carbon dioxide concentration as cues.

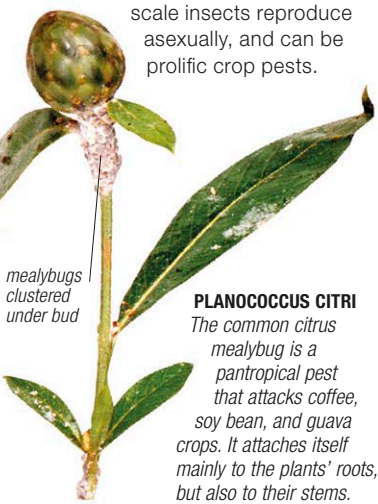
Superfamily Coccoidea

Scale insects

Occurrence 7,000 spp. worldwide, especially in tropical and subtropical regions; on all parts of many plants



Scale insects are so highly adapted to a sap-sucking life that they hardly resemble insects at all. The females are often wingless and legless, and spend their adult lives permanently attached to plants. Most are under $\frac{3}{16}$ in (1 cm) long, with an oval body covered by a smooth or waxy scale. The males look very different, and often have wings. Many scale insects reproduce asexually, and can be prolific crop pests.



Family Fulgoridae

Fulgorids

Occurrence 800 spp. in tropical and subtropical regions; mostly on trees and woody shrubs in well-vegetated areas



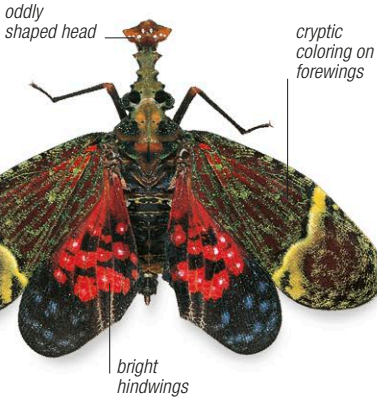
Many tropical fulgorids are notable for their bizarrely shaped, bulbous heads, and are known as lantern bugs. Their eyes are located at the sides of the head, above the antennae. The thorax is often quite large, as are the wings, which have cross veins. These sap-suckers are $\frac{5}{16}$ –4 in (0.8–10 cm) long and have eyespots on the hindwings that can be flashed to deter predators. Adults rest and feed during the day and fly at night. Eggs are laid on host plants and covered by a frothy secretion, which hardens to protect them.



FULGORA LATERNARIA
Also called the lanternfly or peanut-headed bug, this species has a bulbous head that was once thought to glow. It occurs in South America and the West Indies.



FULGORA SERVILLEI
Found in the forests of South America, this species has a very enlarged head with tooth-like markings. Like most species in the family, it is a sap-sucker.



PHRICTUS QUINQUEPARTITUS
The wart-headed bug has colorful wings and an unusual head. It is found in woodland and forests of Panama, Brazil, Colombia, and Costa Rica.

Family Gerridae

Pond skaters

Occurrence 500 spp. worldwide; in water bodies ranging from pools, ponds, streams, rivers, and lakes, to warm oceans



Also known as water-striders, these fast-moving bugs are $\frac{1}{16}$ – $1\frac{1}{2}$ in (0.2–3.5 cm) long, dark brown or black, and covered with short, velvety hairs. The elongated body has short, stout front legs used for grasping prey, and long middle and hindlegs that help to spread its weight over the water film. These predatory bugs detect prey

using ripple-sensitive hairs. Mating follows ripple courtship signals, and eggs are laid on floating objects or embedded in water plants.



GERRIS SP.
These bugs are widespread and live on water surfaces. Ripples made by prey attract the bugs, which move quickly to attack.

Family Hydrometridae

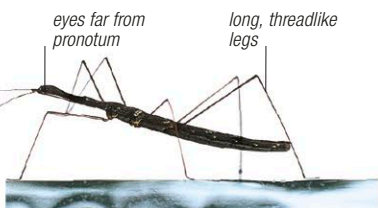
Water measurers

Occurrence 120 spp. worldwide; in standing water bodies, on floating plants



Like pond skaters, water measurers, also called marsh-treaders, live and feed on the surface film, but they move in a slow and stealthy way, creeping up on their prey. They are $\frac{1}{8}$ – $\frac{3}{4}$ in (0.3–2.2 cm) long, with an elongated head and bulging eyes. Most species have either short wings or no wings

at all. Water measurers are predators and scavengers in all life stages, using a spearing and sucking technique to feed; they prefer dead or dying prey.



HYDROMETRA STAGNORUM
Like all water measurers, this European bug is sensitive to surface vibrations and can locate small prey, which it spears with sharp mouthparts.

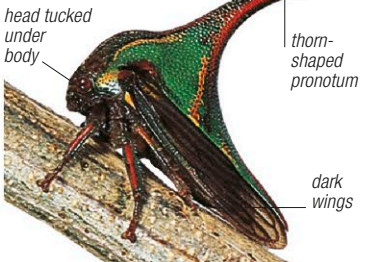
Family Membracidae

Treehoppers

Occurrence 2,500 spp. worldwide, mainly in warmer regions; especially in understory layers in tropical forests



The large, distinctive pronotum, often shaped like a thorn, gives this family the alternative name of thorn bugs. These plant sap-suckers are mostly green, brown, or brightly colored, measuring $\frac{3}{16}$ – $\frac{1}{2}$ in (0.5–1.5 cm) in length. Female treehoppers lay egg masses in slits made in the host plant's bark and, depending on the species, a single female may lay up to 300 eggs. The female frequently guards them until they hatch. Ants protect many species in return for honeydew.



UMBONIA CRASSICORNIS
Members of this genus are found in woodland and forests in North and South America and Southeast Asia. The pronotum is tough and can puncture skin or penetrate shoes.

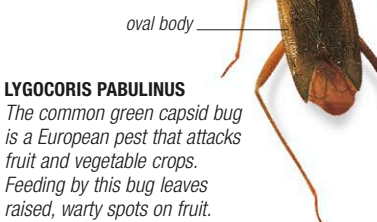
Family Miridae

Plant bugs

Occurrence 8,000 spp. worldwide; on all aerial parts of a wide range of plants, including crops



Also known as capsid bugs, these fragile, elongated or oval insects form the largest family of true bugs. They show a great variety of colors and markings—some resemble ants—and are generally $\frac{1}{16}$ – $\frac{1}{2}$ in (2–16 mm) long. Most are sap-suckers, but some scavenge or hunt live prey. Plant bugs protect themselves by having cryptic coloring, or by giving off a noxious scent when threatened. Eggs are inserted inside plant tissues; a female may lay a total of 30–100 eggs during her adult life. Many species are fruit and crop pests.



LYGOCORIS PABULINUS
The common green capsid bug is a European pest that attacks fruit and vegetable crops. Feeding by this bug leaves raised, warty spots on fruit.